# Sparta data audit (D2): data flow

ANCP Group M. Planned flow for the H&M public dataset; the transcactions file is not loaded yet, so no counts or results are shown.

## 1 Sources

**H&M Personalized Fashion Recommendations (Kaggle)**
articles.csv: product type, colour, section, index group
customers.csv: hashed ID, age, club status
transactions_train.csv: date, customer, article, price, channel

Access: Kaggle account, rules accepted
Licence: competition rules, non-commercial research and education
Status: files pending

**Google Trends (optional)**
Menswear search interest, India, CSV export; joined on week in step 4 if used
Public, aggregated, no personal data
Status: optional

**Sparta pilot app (future)**
Opt-in, pseudonymised wardrobe data
Not available until the pilot runs
Not part of this audit

**Synthetic test data**
Only for testing the cleaning script
Never used for results
Clearly labelled as synthetic

## 2 Raw zone

**Raw files**
Folder: data-audit/raw/
Not committed to git (size and licence)
Recorded for each file:
- source link
- date accessed
- SHA-256 checksum
- row and column counts

**Trends export**
data-audit/raw/trends/
Same records as above

## 3 Audit and cleaning

1. **Schema and type checks**
   dates, IDs, prices, categories
2. **Missing values**
   count per column; drop, fill or flag, why
3. **Duplicates**
   repeated rows and article IDs; keep or merge
4. **Outliers**
   price and age ranges; cap, flag or keep
5. **Menswear filter**
   index group = Menswear
6. **Minimise**
   drop postal code, free text; keep hashed IDs
7. **Bias and representativeness checks**
   age, channel, product mix, season, brand

Each treatment is logged with its reason (clean.py)

## 4 Feature engineering

**Feature engineering log**
Each feature: logic and leakage risk

- week, month, year from date
- season from date (hemisphere)
- colour family from colour name
- category mapped to Sparta taxonomy (FDD section 13)
- price band
- customer age group

Leakage control: features use only information available at the time; a time-based split is used later.

## 5 Outputs

**Cleaned dataset**
version tag data-v1.0
CSV or Parquet

**Data dictionary and checksum**
every column; SHA-256

**Audit report (PDF)**
maximum 8 pages

**Datasheet (appendix)**
3 to 4 pages

**Used by**
D3: EDA notebook
D4: modelling checkpoint
Both read the cleaned dataset

## Controls that apply along the whole flow

- Licence: raw H&M files stay out of git; rules say non-commercial research and education only. Check redistribution before publishing any derivative.
- Privacy: customer IDs are already hashed; no attempt to re-identify; postal code and free text are dropped; no personal data in the repository (DPDP Act, 2023 statement in the report).
- Provenance: source link, date accessed and SHA-256 checksum recorded for every raw file; the cleaning script reproduces the cleaned dataset from raw data.
- Scope: H&M is one brand's mostly European customers, so it is a stand-in for purchase demand, not for Sparta users. Representativeness checks say so.
- Dashed boxes are not used in this audit: they are optional or not yet available.